Day 2 — Fluid Mechanics I

Comprehensive Problem Set

June 18, 2026 • Concept Review + 20 Questions

Focus: Fluid statics & pressure, the continuity equation ($Q = VA$), the Bernoulli/energy equation, Reynolds number, and laminar vs. turbulent flow.

How to use this set: Start with the *Concept Review* to refresh the key ideas and formulas, then work the problems using only the *NCEES FE Reference Handbook*. Use $g = 9.81$ m/s^2, $\rho_{water} = 1000$ kg/m^3, and $\gamma_{water} = 9.81$ kN/m^3 unless stated otherwise. Choose the answer *nearest* to your result. Two items are numeric-entry. Write units at every step.

Concept Review — Fluid Mechanics I

Read this first. Every relation below appears in the NCEES FE Reference Handbook (Fluid Mechanics section); the goal is to recognize and locate each one quickly.

1. Fluid Properties

- **Density / specific weight:** $\gamma = \rho g$. Water: $\rho = 1000$ kg/m^3, $\gamma = 9.81$ kN/m^3 = 9810 N/m^3.
- **Specific gravity:** SG $= \dfrac{\rho}{\rho_{water}} = \dfrac{\gamma}{\gamma_{water}}$ (dimensionless); multiply by γ_{water} to recover a fluid's γ.
- **Viscosity:** dynamic μ (Pa·s); kinematic $\nu = \dfrac{\mu}{\rho}$ (m^2/s).
- **Newtonian fluid:** $\tau = \mu \dfrac{dv}{dy}$ — shear stress is *linearly* proportional to the velocity gradient (rate of shear strain), with μ constant.

2. Fluid Statics

- **Hydrostatic pressure (gauge):** $p = \rho g h = \gamma h$; pressure rises linearly with depth, $\dfrac{dp}{dz} = -\gamma$.
- **Absolute vs. gauge:** $p_{abs} = p_{atm} + p_{gauge}$, with $p_{atm} \approx 101.3$ kPa = 760 mm Hg = 10.33 m of water.
- **Pressure head:** $h = \dfrac{p}{\gamma}$ (height of a fluid column giving pressure p).
- **Manometry:** step through the tube — *add* γh moving down, *subtract* moving up; equate the two ends.
- **Force on a submerged plane:** $F_R = \gamma \bar{h} A$, where $\bar{h}$ is the depth to the area's centroid.
- **Center of pressure:** $y_{cp} = \bar{y} + \dfrac{I_{xc}}{\bar{y} A}$, with $I_{xc} = \dfrac{bh^3}{12}$ (rectangle) or $\dfrac{\pi D^4}{64}$ (circle). It always lies *below* the centroid.
- **Buoyancy (Archimedes):** $F_B = \gamma_{fluid} V_{disp}$; apparent (submerged) weight $= W_{air} - F_B$.

3. Continuity (Conservation of Mass)

- **Volumetric flow:** $Q = AV$; circular pipe area $A = \frac{\pi}{4}D^2$.
- **Mass flow:** $\dot{m} = \rho Q = \rho AV$.
- **Incompressible, steady:** $A_1V_1 = A_2V_2 \;\Rightarrow\; V_2 = V_1\left(\frac{D_1}{D_2}\right)^2$ — velocity scales with the *square* of the diameter ratio.

4. Energy — the Bernoulli Equation

- **Bernoulli** (steady, incompressible, no losses, along a streamline):

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2.$$

  The three terms are the *pressure head*, *velocity head*, and *elevation head*; their sum is the total head H.

- **Extended energy equation** (with machines and losses):

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + h_{pump} = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + h_{turbine} + h_L.$$

- **Torricelli** (free discharge a depth h below the surface): $v = \sqrt{2gh}$.
- **Pitot tube:** $V = \sqrt{2g\,\Delta h} = \sqrt{\frac{2\,\Delta p}{\rho}}$ (stagnation minus static pressure).

5. Reynolds Number & Flow Regime

- **Reynolds number:** $Re = \frac{\rho VD}{\mu} = \frac{VD}{\nu}$ (ratio of inertial to viscous forces). For non-circular sections use the hydraulic diameter $D_H = \frac{4A}{P}$.
- **Regimes (pipe flow):** laminar $Re < 2100$; transitional $2100-4000$; turbulent $Re > 4000$.
- **Laminar friction factor:** $f = \frac{64}{Re}$ (Darcy). Turbulent f comes from the Moody diagram / Colebrook equation.

Handbook: Fluid Mechanics — properties, statics, continuity, energy/Bernoulli, Reynolds number

Questions

1. The gauge pressure at the bottom of a 6.0 m deep, open tank of water is most nearly:

   (A) 5.89 kPa
   (B) 58.9 kPa
   (C) 588 kPa
   (D) 58.9 Pa

2. A tank holds a liquid of specific gravity 0.85 to a depth of 4.0 m. The gauge pressure at the bottom is most nearly:

   (A) 28.4 kPa
   (B) 33.4 kPa
   (C) 39.2 kPa
   (D) 47.1 kPa

3. A gauge pressure of 200 kPa is equivalent to what height of a water column?

   (A) 14.7 m
   (B) 20.4 m
   (C) 2.04 m
   (D) 200 m

4. Standard atmospheric pressure (101.3 kPa) is equivalent to what height of a mercury column (SG = 13.6)?

   (A) 760 mm
   (B) 1013 mm
   (C) 101 mm
   (D) 13.6 mm

5. At a site where the local atmospheric (barometric) pressure is 98 kPa, a Bourdon gauge reads 150 kPa. The absolute pressure is:

   (A) 52 kPa
   (B) 150 kPa
   (C) 248 kPa
   (D) 198 kPa

6. A vertical rectangular gate is 2.0 m wide and 3.0 m tall, with its top edge at the water surface. The resultant hydrostatic force on the gate is most nearly:

   (A) 29.4 kN
   (B) 44.1 kN
   (C) 88.3 kN
   (D) 176.6 kN

7. For the gate of Problem 6, the center of pressure lies at what depth below the free surface?

   (A) 1.50 m
   (B) 2.00 m
   (C) 2.25 m
   (D) 3.00 m

8. A solid object weighs 50 N in air and 30 N when fully submerged in water. The volume of the object is most nearly:

   (A) 2.04 L
   (B) 1.02 L
   (C) 5.10 L
   (D) 20.4 L

9. **(Numeric entry)** Water flows through a pipe of inside diameter 100 mm at a mean velocity of 2.5 m/s. Determine the volumetric flow rate, in L/s.

10. Water flows steadily from a 200 mm pipe into a 100 mm pipe. If the velocity in the 200 mm pipe is 1.5 m/s, the velocity in the 100 mm pipe is:

(A) 0.375 m/s (B) 3.0 m/s
(C) 6.0 m/s (D) 12.0 m/s

11. Water ($\rho = 1000$ kg/m^3) flows at a volumetric rate of 0.05 m^3/s. The corresponding mass flow rate is:

(A) 0.05 kg/s (B) 5 kg/s
(C) 50 kg/s (D) 500 kg/s

12. Water discharges through a small orifice located 4.0 m below the free surface of a large open tank. Neglecting losses, the velocity of the jet is most nearly:

(A) 4.43 m/s (B) 6.26 m/s
(C) 8.86 m/s (D) 39.2 m/s

13. A pitot tube placed in a water flow registers a stagnation-minus-static pressure difference equal to 0.50 m of water. The flow velocity is most nearly:

(A) 2.21 m/s (B) 3.13 m/s
(C) 4.43 m/s (D) 9.81 m/s

14. Water flows in a horizontal pipe. At point 1 the velocity is 2.0 m/s and the pressure is 300 kPa. The pipe then narrows so that the velocity at point 2 is 6.0 m/s. Neglecting losses, the pressure at point 2 is most nearly:

(A) 268 kPa (B) 284 kPa
(C) 300 kPa (D) 316 kPa

15. Water flows up a pipe at a steady 3.0 m/s. At point 1 (elevation 0) the pressure is 200 kPa; point 2 is 5.0 m higher with the same velocity. Neglecting losses, the pressure at point 2 is most nearly:

(A) 151 kPa (B) 200 kPa
(C) 249 kPa (D) 49 kPa

16. **(Numeric entry)** Water ($\rho = 1000$ kg/m^3, $\mu = 1.0 \times 10^{-3}$ Pa $\cdot$ s) flows at 2.0 m/s in a 50 mm diameter pipe. Compute the Reynolds number.

17. A fluid with kinematic viscosity $\nu = 1.0 \times 10^{-6}$ m^2/s flows at 0.80 m/s in a 25 mm diameter pipe. The Reynolds number is:

(A) 5,000 (B) 20,000
(C) 50,000 (D) 200,000

18. Pipe flow is generally considered *laminar* when the Reynolds number is below approximately:

(A) 2,100 (B) 4,000
(C) 21,000 (D) 640

19. For fully developed laminar pipe flow at $Re = 1600$, the Darcy friction factor is:

(A) 0.016
(B) 0.025
(C) 0.040
(D) 0.400

20. Which statement best characterizes a *Newtonian* fluid?

(A) Shear stress is proportional to the velocity gradient (rate of shear strain)
(B) Viscosity increases as the shear rate increases
(C) Shear stress is independent of the velocity gradient
(D) It has zero viscosity at all shear rates

Answer Key

Q	Ans	Q	Ans	Q	Ans
1	B	8	A	15	A
2	B	9	19.6 L/s	16	1.0×10^5
3	B	10	C	17	B
4	A	11	C	18	A
5	C	12	C	19	C
6	C	13	B	20	A
7	B	14	B		

Worked Solutions

1. $p = \rho g h = (1000)(9.81)(6.0) = 58{,}860 \text{ Pa} \approx \mathbf{58.9\ kPa}$.

   Handbook: Fluid Mechanics — Pressure in a static liquid, $p = \rho g h$

2. $p = \text{SG}\,\gamma_w h = (0.85)(9.81 \text{ kN/m}^3)(4.0) = \mathbf{33.4\ kPa}$.

   Handbook: Fluid Properties — specific gravity; $p = \gamma h$

3. $h = \dfrac{p}{\gamma} = \dfrac{200{,}000}{9810} = \mathbf{20.4\ m}$ of water.

   Handbook: Fluid Mechanics — pressure head, $h = p/\gamma$

4. $h = \dfrac{p}{\text{SG}\,\gamma_w} = \dfrac{101{,}300}{(13.6)(9810)} = 0.759 \text{ m} \approx \mathbf{760\ mm\ Hg}$.

   Handbook: Fluid Mechanics — manometry, $p = \rho g h$

5. Absolute = atmospheric + gauge = $98 + 150 = \mathbf{248\ kPa}$.

   Handbook: Fluid Mechanics — absolute vs. gauge pressure

6. $F_R = \rho g\, h_c A$, with centroid depth $h_c = 1.5$ m and $A = (2)(3) = 6 \text{ m}^2$:

$$F_R = (1000)(9.81)(1.5)(6) = 88{,}290 \text{ N} \approx \mathbf{88.3\ kN}.$$

   Handbook: Forces on Submerged Surfaces / Center of Pressure

7. $y_{cp} = y_c + \dfrac{I_{xc}}{y_c A}$, with $I_{xc} = \dfrac{bh^3}{12} = \dfrac{(2)(3)^3}{12} = 4.5 \text{ m}^4$:

$$y_{cp} = 1.5 + \frac{4.5}{(1.5)(6)} = 1.5 + 0.5 = \mathbf{2.00\ m}.$$

   Handbook: Forces on Submerged Surfaces / Center of Pressure

8. Buoyant force $F_B = W_{air} - W_{sub} = 50 - 30 = 20$ N. Since $F_B = \rho g V$,

$$V = \frac{20}{(1000)(9.81)} = 2.04 \times 10^{-3}\ \text{m}^3 = \mathbf{2.04\ L}.$$

Handbook: Fluid Mechanics — Archimedes' principle / buoyancy

9. $A = \frac{\pi}{4}D^2 = \frac{\pi}{4}(0.10)^2 = 7.854 \times 10^{-3}\ \text{m}^2$;

$$Q = VA = (2.5)(7.854 \times 10^{-3}) = 0.0196\ \text{m}^3/\text{s} = \mathbf{19.6\ L/s}.$$

Handbook: Continuity Equation, $Q = AV$

10. Continuity $A_1V_1 = A_2V_2 \Rightarrow V_2 = V_1\left(\frac{D_1}{D_2}\right)^2 = (1.5)\left(\frac{200}{100}\right)^2 = (1.5)(4) = \mathbf{6.0\ m/s}$.

Handbook: Continuity Equation, $A_1V_1 = A_2V_2$

11. $\dot{m} = \rho Q = (1000)(0.05) = \mathbf{50\ kg/s}$.

Handbook: Continuity — mass flow rate, $\dot{m} = \rho Q$

12. Torricelli (Bernoulli with free surfaces): $v = \sqrt{2gh} = \sqrt{2(9.81)(4.0)} = \sqrt{78.48} = \mathbf{8.86\ m/s}$.

Handbook: Bernoulli/Energy Equation — orifice discharge

13. Pitot tube: $v = \sqrt{2g\,\Delta h} = \sqrt{2(9.81)(0.50)} = \sqrt{9.81} = \mathbf{3.13\ m/s}$.

Handbook: Fluid Mechanics — pitot tube / stagnation pressure

14. Horizontal Bernoulli: $p_2 = p_1 + \frac{1}{2}\rho\left(V_1^2 - V_2^2\right) = 300{,}000 + \frac{1}{2}(1000)(2^2 - 6^2)$

$$= 300{,}000 - 16{,}000 = 284{,}000\ \text{Pa} = \mathbf{284\ kPa}.$$

Handbook: Bernoulli Equation (energy along a streamline)

15. Same velocity, so the kinetic terms cancel: $p_2 = p_1 - \rho g(z_2 - z_1) = 200{,}000 - (1000)(9.81)(5.0)$

$$= 200{,}000 - 49{,}050 = 150{,}950\ \text{Pa} \approx \mathbf{151\ kPa}.$$

Handbook: Bernoulli/Energy Equation — elevation (pressure) head

16. $Re = \frac{\rho V D}{\mu} = \frac{(1000)(2.0)(0.050)}{1.0 \times 10^{-3}} = \mathbf{1.0 \times 10^5}$. (Turbulent, since $Re > 4000$.)

Handbook: Reynolds Number, $Re = \rho V D/\mu$

17. $Re = \frac{VD}{\nu} = \frac{(0.80)(0.025)}{1.0 \times 10^{-6}} = \mathbf{20{,}000}$.

Handbook: Reynolds Number, $Re = VD/\nu$

18. By convention, pipe flow is laminar for $Re \lesssim$ **2100**, transitional for $2100 - 4000$, and turbulent above ≈ 4000.

Handbook: Reynolds Number — flow regimes

19. Laminar friction factor: $f = \dfrac{64}{Re} = \dfrac{64}{1600} = \mathbf{0.040}$.

Handbook: Head Loss / Darcy friction factor, $f = 64/Re$ (laminar)

20. A Newtonian fluid has shear stress linearly proportional to the velocity gradient, $\tau = \mu \dfrac{dv}{dy}$, with constant μ. **Answer (A).**

Handbook: Fluid Mechanics — Newtonian fluid / viscosity

All numeric results independently verified. Keep practicing locating each equation in the NCEES FE Reference Handbook.